Marco Mistretta

PhD Student in Artificial Intelligence
ex Amazon Applied Scientist Intern

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Professional Summary

- PhD researcher in **AI** at MICC, University of Florence (Italy), under **Prof. Andrew D. Bagdanov**
- Top conferences Publications: **CVPR 2026, ICLR 2026, ICLR 2025, ECCV 2024**
- Former **Applied Scientist Intern at Amazon (London)**.

Professional Experience

Jul 2025 – Dec 2025 **Applied Scientist Intern**, *Amazon Rufus Team*, London, UK
- Fine-tuned, evaluated, and deployed MLLMs impacting millions of customers.

Education

2023–Present **PhD in Artificial Intelligence**, *University of Florence, Italy*
Multimodal Vision-Language Models, Incremental & Prompt Learning
Supervisors: A. D. Bagdanov, M. Bertini

2021–2023 **MSc in Artificial Intelligence**, *University of Florence, Italy*
Thesis: *RE-Tune — Incremental Fine-Tuning of Biomedical VLMs*, Grade: 110/110L

2018–2021 **BSc in Computer Engineering**, *University of Florence, Italy*
Thesis: *AI-driven Scarlatti Sonatas Generation*, Grade: 105/110

Selected Publications

- "*IsoCLIP: Decomposing CLIP Projectors for Efficient Intra-modal Alignment*." **(CVPR 2026)**
  Magistri S., Goswami D., **Mistretta M.**, Twardowski B., van de Weijer J., Bagdanov A. D.
  Reveals how CLIP projectors induce inter- and intra-modal operators causing intra-modal misalignment.
- "*SpectralGCD: Spectral Concept Selection and Cross-modal Representation Learning for Generalized Category Discovery*." **(ICLR 2026)**
  Caselli L., **Mistretta M.**, Magistri S., Bagdanov A. D.
  Multimodal approach for GCD via CLIP-based spectral concept selection. [code] [paper]
- "*Cross the Gap: Exposing the Intra-modal Misalignment in CLIP via Modality Inversion*." **(ICLR 2025)**
  Mistretta M.*, Baldrati A.*, Agnolucci L.*, Bertini M., Bagdanov A. D.
  Exposes intra-modal misalignment in CLIP via a modality inversion based analysis. [code] [paper]
- "*Improving Zero-shot Generalization of Learned Prompts via Unsupervised Knowledge Distillation*." **(ECCV 2024)**
  Mistretta M.*, Baldrati A.*, Bertini M., Bagdanov A. D.
  Add-on for prompt learning techniques that improves prompt generalization without labels. [code] [paper]

Research Interests

- **Multimodal Learning**: Combining vision and language for richer model understanding.
- **Prompt Learning**: Optimizing prompts to effectively guide AI models for improved task performance.
- **Few-Shot Learning**: Adapting AI to new tasks with limited examples.
- **Incremental Learning**: Enabling models to continuously learn without forgetting previous knowledge.
- **Contrastive Self-Supervised Learning**: Extracting insights from unlabeled data.

Reviewer Roles

- **Conferences**: NeurIPS 2024, ICLR 2025, NeurIPS 2025, CVPR 2026
- **Journal**: IEEE Transactions on Multimedia/Circuits and Systems for Video Technology

Teaching and Mentoring

Jan 2024 – Jan 2026 **Teaching Assistant**, *University of Florence*
- Delivering interactive lessons on C/C++ and Python to over 200 bachelor students.

Jan 2020 – Nov 2020 **Student Ambassador**, *University of Florence*
- Mentoring students on exams projects, internships, and career development.

Technical Skills

Programming	Python, Java, C++, MATLAB, R
Frameworks	PyTorch, TensorFlow, NumPy, OpenCV
Miscellanous	Git, Docker, Bash

Personal Interests

Music
- **Flute Player**: Played transverse flute for 10 years in local bands and orchestras.
- **Chorister** at "Coro Giovanile Effetti Sonori" (National Competition Winner)
- **Guitarist and Songwriter** for "I Green Clouds" (Provincial Competition Winner)

Gaming: Enjoying strategic and cooperative games that enhance problem-solving and teamwork.

Languages

English	Professional working proficiency
Italian	Native